

DevOps Engineer

Step 7: Deploy Production Projects

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

Production deployment is a system, not a button. You need repeatable infrastructure, configuration and secret management, health checks, logging, monitoring, backups, rollback and a clear incident process.

AWS guidance recommends fully automated integration/deployment, multiple environments, small reversible changes and safe rollout strategies such as rolling, canary and blue/green deployments. ■cite■turn0search13■turn0search16■

Core Concepts — Detailed Breakdown

- Configuration: separate environment-specific settings from code.
- Secrets: store credentials in a secret manager or protected environment; never bake them into images.
- Health: readiness/health endpoints and dependency checks.
- Observability: metrics, logs and traces; track latency, errors, traffic and resource saturation.
- Reliability: backups, restore tests, redundancy and sensible timeouts.
- Deployment strategies: rolling, canary, blue/green and feature flags.
- Rollback: know exactly which version to restore and how to verify recovery.
- Incident response: detect → assess → mitigate → communicate → recover → learn.
- Security: patch dependencies, restrict access, scan images/IaC and audit changes.

What You Should Be Able To Do

- Deploy with CI/CD.
- Add HTTPS.
- Add health checks.
- Centralize logs.
- Create at least one alert.
- Perform a rollback drill.
- Write a short incident runbook.

Hands-on Project / Practice

- Deploy a containerized application with staging and production environments. Add CI/CD, HTTPS, health checks, monitoring, backups and a rollback plan. Intentionally deploy a bad version and practice recovery.

Recommended References

- Terraform Documentation — <https://developer.hashicorp.com/terraform/docs>
- GitHub Actions Documentation — <https://docs.github.com/en/actions>
- AWS Well-Architected Framework — <https://docs.aws.amazon.com/wellarchitected/latest/framework/>

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.